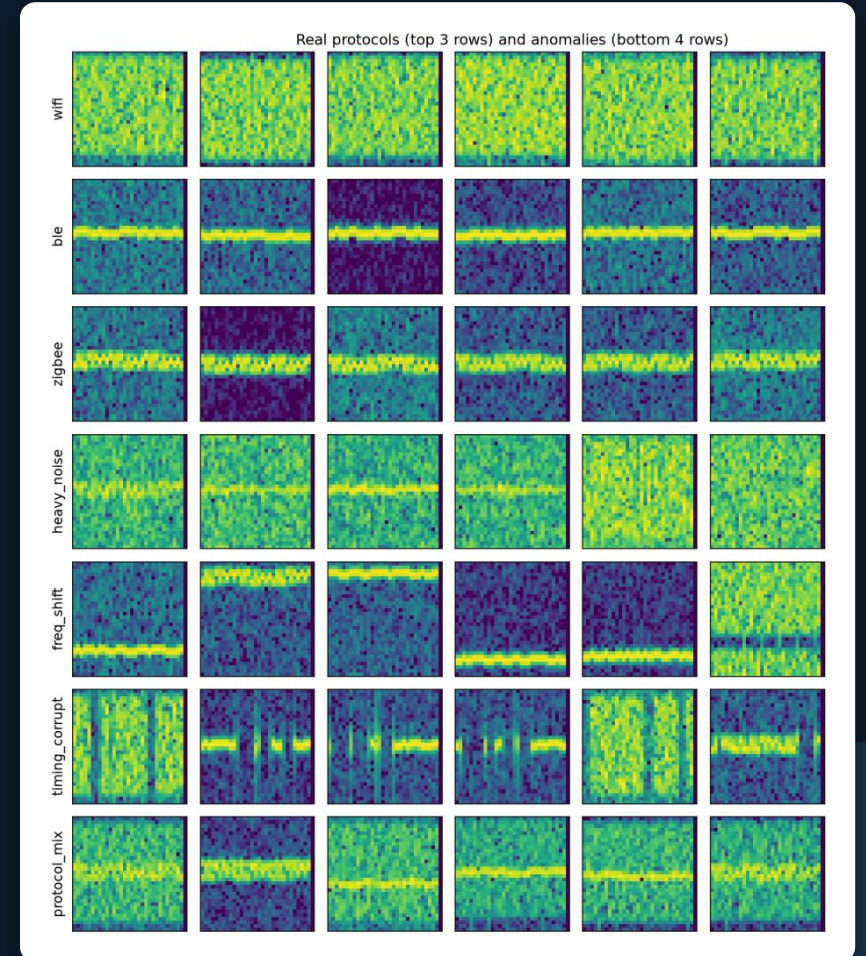


Deep Generative Models for Synthetic RF Signals & Anomaly Detection

Comparing a **VAE**, a **Flow**, a **Diffusion Model**, and **Mamba**

Tristan Ruel · Student No. 1012867710

University of Toronto



3 protocols (top) · 4 anomaly types (bottom)

The Problem & The Data

Realistic RF datasets are hard to collect, label, and balance — corrupted, malformed, and protocol-inconsistent transmissions are rare. Can a model learn the distribution of valid signals well enough to both generate realistic samples and flag ones that do not fit?

1

Synthetic, reproducible corpus

Wi-Fi (OFDM), BLE (GFSK), ZigBee (O-QPSK/DSSS) — pure-NumPy I/Q, seeded.

2

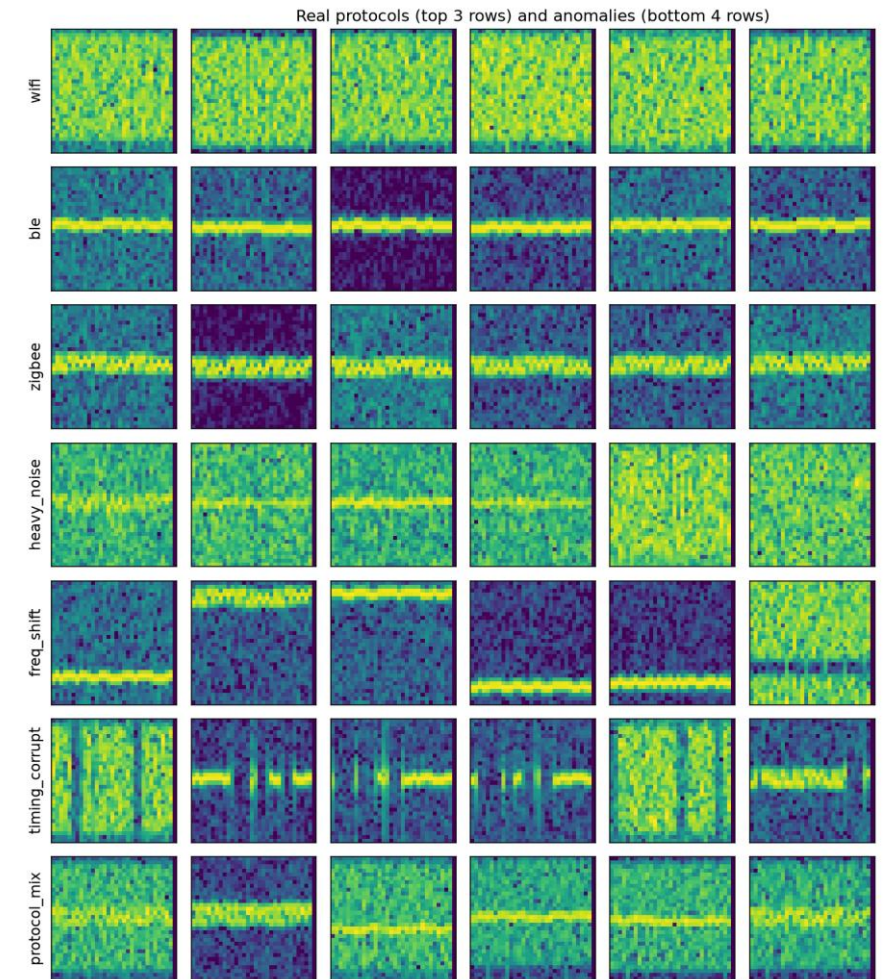
Shared 32×32 spectrograms

One log-STFT representation for every model → a fair comparison.

3

Four anomaly types (test-only)

Heavy noise · frequency shift · timing corruption · protocol mixing.



Four Generative Models

All class-conditional. Anomaly score = min over classes of each model's native score.



Conditional VAE

LATENT-VARIABLE

Conv encoder–decoder;
maximise the ELBO.

score: neg. ELBO



RealNVP Flow

EXPLICIT LIKELIHOOD

Logit + 8 coupling layers; exact
 $\log p(x)$.

score: neg. log-likelihood



Conditional DDPM

DENOISING / DIFFUSION

Class-cond. U-Net predicts
added noise.

score: denoising error



Mamba (I/Q)

AUTOREGRESSIVE SSM

Selective SSM over raw I/Q
tokens; exact discrete $p(x)$.

score: neg. log-likelihood

Inputs: VAE / flow / DDPM share a peak-normalised 32×32 STFT spectrogram; Mamba is autoregressive over the raw I/Q sequence. All train on **normal signals only**; anomalies are held out.

Results

Anomaly-detection ROC-AUC (trained on normal signals only)

● **0.983**

RealNVP flow

explicit likelihood — best detector

● **0.972**

Conditional DDPM

denoising — best fidelity (Fréchet 0.16)

● **0.846**

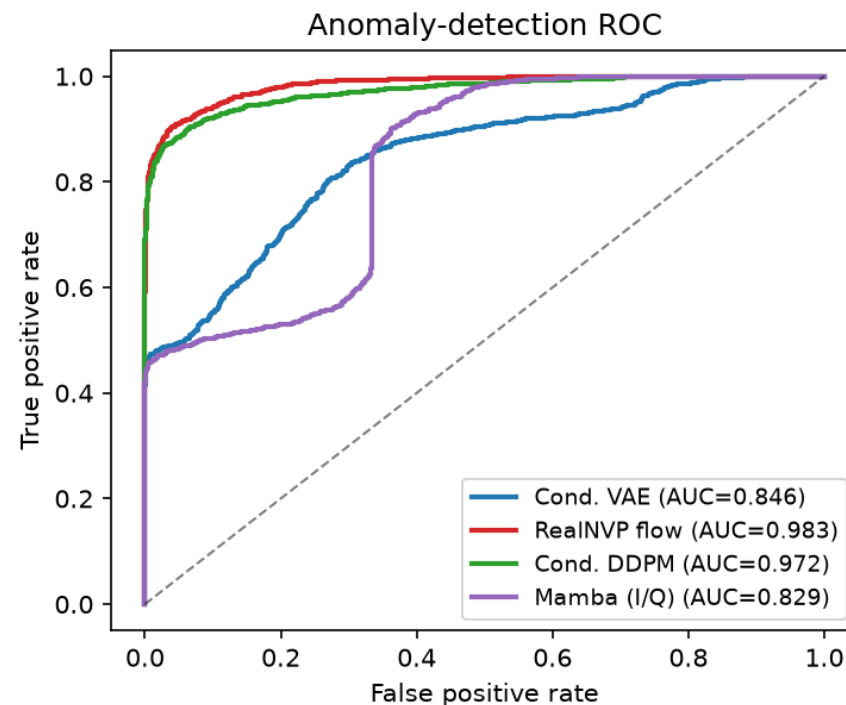
Conditional VAE

latent — cheapest, but over-smooths

● **0.829**

Mamba (raw I/Q)

autoregressive — clean discrete likelihood



Class-conditional generation succeeded for all four (100% classifier accuracy).

Conclusion

For RF anomaly detection under limited data, prefer likelihood-based or denoising models over the VAE.



Flow

Best detector (AUC 0.983) and exact likelihoods; largest model.



Diffusion

Near-best detection (0.972) + most realistic samples; slow sampling.



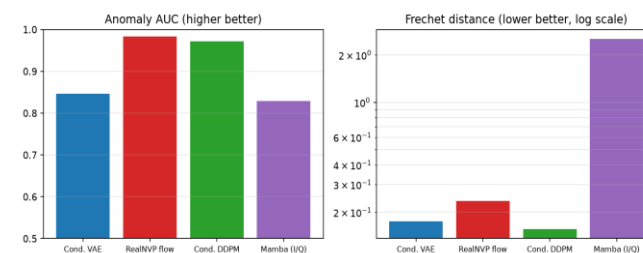
VAE

Cheapest & fastest with an encoder; weak detector, blurry samples.



Mamba (I/Q)

Raw-I/Q autoregressive; clean discrete likelihood, but weakest detector & costliest.



Anomaly AUC ↑ · Fréchet ↓

github → reproducible pipeline · report · demo notebook

Future: raw-I/Q flows · over-the-air capture · flow-detector + diffusion-generator hybrid